as per syllabus

- projectile plotting
- raycasting
- soft body dynamics

Raycastas and area detection/simple ai

Physics basics

Forward Eulors method

eulors method for the equations of motion

$$x_{t+1} = x_t + v_t$$

$$v_{t+1} = v_t + a_t$$

we use the semi implicite eulers method where we calculate the velocity on the next step first, this prevents us from getting infinite energy in spring systems

in a regular euler system, when it samples energy from the first step, it samples the current velocity at it's highest, and so when the next step happens and the velocity updates

it also

$$x_{t+1} = x_t + v_{t+1}$$

$$x_{t+1} = x_t + v_t + a_t$$

math behinds soft body physics

Differenc between solid and point masses

Point masses

mass position velocity

Shapes

list of point masses in order

Collision detection

Check overlap for points if inside

Raycasting

for point detection in horizontal lines

get closest edge, detect and resolve collisions

Springs

for shapes

force that keeps two objects at a certain distance

- rest distance
- force is applied

maintaining shape pressure

shape matching

pin joints and spring joints

soft body plugin godot